

AI Operating System for Small Businesses

Complete Project Blueprint

Vision

Build an AI-powered SaaS platform that becomes the central intelligence layer for small businesses. Instead of switching between Gmail, Drive, PDFs, Calendar and tasks, users ask one AI assistant that can search, summarize, automate and execute workflows.

Problem

Business information is scattered across many apps, making searching, reporting and decision making slow.

Target Users

Small businesses (5-50 employees), agencies, startups, consultancies, clinics, schools.

Core Modules

Authentication; Company Dashboard; AI Chat; Document Knowledge Base (RAG); Gmail Integration; Google Drive Integration; Calendar; Task Manager; Workflow Automation; Analytics; Team Management.

AI Architecture

Next.js frontend -> FastAPI backend -> LangGraph orchestrator -> specialized agents (Search, Email, Calendar, Task, Document) -> Qdrant vector DB -> PostgreSQL -> Gemini/Groq LLM.

Recommended Tech Stack

Frontend: Next.js, React, Tailwind CSS. Backend: FastAPI, Python. Database: PostgreSQL. Vector DB: Qdrant. Auth/Storage: Supabase Free. AI: Gemini API free tier + Sentence Transformers. Automation: n8n Community. Parsing: PyMuPDF, python-docx, pandas/openpyxl. OCR: Tesseract. Deployment: Docker, GitHub, Vercel(frontend), Railway/Render/local for backend during development.

Development Roadmap

Phase 1: Auth, dashboard, document upload, RAG chat. Phase 2: Gmail + Drive + Calendar. Phase 3: AI agents, task extraction, automations. Phase 4: Analytics, notifications, team collaboration. Phase 5: Billing, public SaaS launch.

Suggested Folder Structure

frontend/, backend/, ai-services/, agents/, rag/, workflows/, docker/, docs/.

Resume Highlights

Multi-agent architecture, production RAG, SaaS, multi-tenancy, AI automation, Docker deployment, analytics, secure authentication.

Cost

Development can be done almost entirely for free using free tiers. Paid hosting or API upgrades are only needed when real usage grows.